

ESAT Mathematics 1

SPEED, RECALL & PROBLEM-SOLVING GUIDE

Speed, Recall & Problem-Solving Guide - Revised

Revised edition — real typeset formulas, a fixed 3-tier priority system,
and additional speed tricks folded in

Basis and scope

Syllabus content and terminology are based on the official UAT-UK ESAT Guide Mathematics 1, June 2026.

The priority rankings, shortcuts and speed tactics in this revised document are strategic synthesis, not official UAT-UK topic weighting or claims about question frequency.

The aim is not to teach each topic from scratch. It is to make standard mathematics automatic, reduce unnecessary working, and improve method selection under time pressure.

Source: UAT-UK, ESAT Guide Mathematics 1, June 2026

How to use this guide

Revision note: the priority tiers below are relabelled from the original P1/P2/P3 numbering to AUTOMATE / FLUENT / TRAP-GUARD. The original numbering collided with the syllabus's own unit codes (e.g. Physics unit "P3 Mechanics" vs. priority tag "P1" for that same row) — the new labels mean the same thing but can never be confused with a unit code.

AUTOMATE. Instant recognition, zero hesitation.

FLUENT. A few written steps, but no uncertainty about method.

TRAP-GUARD. Familiar content where avoidable marks are lost.

- Speed trick. A legitimate mathematical shortcut or method-selection rule. Use it only when its conditions are satisfied.
- Memorise cold. Facts, identities, formulae or recognition patterns that should not consume thinking time in the test.

Overall preparation priorities

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	M2 Number	Exact arithmetic, cancellation, fractions, powers, surds, bounds and estimation underpin almost every other unit.
AUTOMATE	M4 Algebra	The main performance gain is selecting the shortest valid method rather than carrying out long routine algebra.
AUTOMATE	M5 Geometry	Fast diagram annotation and theorem recognition can turn multi-step problems into short chains of deductions.
AUTOMATE	M7 Probability	Representation choice, complements and conditional denominators often decide whether a question is short or long.
FLUENT	M3 Ratio & proportion	Multipliers, scale factors and direct/inverse proportion recur in many contexts.
FLUENT	M6 Statistics	The calculations are usually modest; the danger is reading the wrong quantity from a graph or using the wrong denominator.
TRAP-GUARD	M1 Units	Low conceptual difficulty for a strong student, but conversions must be automatic because unit slips are avoidable losses.

Each unit below follows the same shape: Priority ranking, Memorise cold, Ranked speed tricks (plus any additional tricks folded in for this revision, clearly marked), Common time-wasters and traps, and a 10-second pre-answer check.

M1. Units

Official guide: pp. 5-12 | Sections M1.1-M1.2

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Make conversions algebraic and automatic. A strong student should spend almost no reasoning time on standard unit conversion; attention should go to compound units and powers of conversion factors.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Compound-unit conversion	Rates, density and pressure combine numerator and denominator units, so a single missed factor can create errors of 10^3 or 10^6 .
AUTOMATE	Area and volume conversions	The linear conversion factor must be squared or cubed.
FLUENT	Time and speed conversions	Minutes/hours and m/s versus km/h are frequent sources of unnecessary working.
TRAP-GUARD	Standard unit recall	Basic mass, length, capacity and time units should already be automatic.

MEMORISE COLD

- 1 km = 1000 m; 1 m = 100 cm = 1000 mm.
- $1 \text{ m}^2 = 10,000 \text{ cm}^2$; $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$.
- 1 litre = 1000 ml = 1000 cm^3 ; 1 ml = 1 cm^3 ; 1000 litres = 1 m^3 .
- 1 kg = 1000 g; 1 g = 1000 mg.
- 60 s = 1 min; 60 min = 1 h; 24 h = 1 day.
- Useful derived shortcut: 1 m/s = 3.6 km/h; therefore divide km/h by 3.6 to get m/s.
- Useful density shortcut: $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$.

RANKED SPEED TRICKS

1. Treat units like algebra. Write a compound quantity as a fraction. For density 20 g/cm^3 , think $20 \text{ g} / 1 \text{ cm}^3$ and convert numerator and denominator separately.
2. Power the conversion factor. If $1 \text{ m} = 100 \text{ cm}$, then $1 \text{ m}^2 = 100^2 \text{ cm}^2$ and $1 \text{ m}^3 = 100^3 \text{ cm}^3$. Do not memorise every area/volume conversion independently.
3. Convert only what the final unit requires. If the answer is required in km/h, convert distance to km and time to hours before dividing.
4. Use dimensional direction as a check. Changing cm^3 in a denominator to m^3 should make the numerical density much larger, not smaller.
5. Exploit familiar ratios. 30 min = 0.5 h, 15 min = 0.25 h, 45 min = 0.75 h. These avoid unnecessary fraction-to-decimal work.
6. Estimate before finalising. If 35 km is travelled in half an hour, the answer must be about 70 km/h. A result such as 7 or 700 signals a conversion error.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Multiplying by 100 rather than 10,000 when converting m^2 to cm^2 .
- Forgetting that a denominator conversion reverses the numerical effect.

- Mixing minutes and hours inside speed/rate formulae.
- Writing a correct number with an incompatible unit.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Are all component units compatible?
- Did an area/volume conversion need a square/cube?
- Does the numerical direction of the conversion make sense?
- Are the final units exactly those requested?

EXPANDED PATTERN BANK

Mini speed drill - aim for mental conversion

- 72 km/h $\rightarrow$ 20 m/s.
- 2.4 g/cm³ $\rightarrow$ 2400 kg/m³.
- 0.35 m² $\rightarrow$ 3500 cm².
- 4500 cm³ $\rightarrow$ 4.5 litres.
- 36,000 m in 30 min $\rightarrow$ 72 km/h.
- If an answer differs by a factor of 10, 100, 1000 or 1,000,000, suspect the unit conversion before redoing the whole problem.

M2. Number

Official guide: pp. 13-73 | Sections M2.1-M2.14

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Turn arithmetic into pattern recognition. The official guide spans ordering, operations, factors/primes, cancellation, listing, indices, standard form, fractions/decimals/percentages, surds, bounds, rounding and estimation. For a strong student, the main target is to avoid doing arithmetic that simplification can remove.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Cancellation and exact arithmetic	Simplify before evaluation. This reduces workload in fractions, algebra and later physics calculations.
AUTOMATE	Fractions, indices and surds	Exact forms should remain exact until there is a reason to approximate.
AUTOMATE	Bounds and error intervals	A boundary mistake can invalidate otherwise correct arithmetic.
AUTOMATE	Estimation and magnitude checks	Fast estimation catches decimal-place and standard-form errors.
FLUENT	Prime factors, HCF and LCM	Useful directly and as a simplification tool for roots and fractions.
FLUENT	Standard form	Usually easy if coefficients and powers of ten are handled separately.
FLUENT	Systematic counting/listing	Useful bridge into probability; complements often shorten the work.
TRAP-GUARD	Ordering and place value	Should be automatic, but mixed fractions/decimals can still invite needless conversion.

MEMORISE COLD

- BIDMAS/PEMDAS: brackets; indices (including roots/reciprocals); division and multiplication left-to-right; addition and subtraction left-to-right.
- Prime numbers below 30: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29.
- Divisibility: 2 last digit even; 3 digit sum divisible by 3; 4 last two digits; 5 ends 0/5; 6 divisible by 2 and 3; 7 subtract twice the last digit from the remaining digits and repeat; 8 last three digits; 9 digit sum divisible by 9.
- For two positive integers: $\text{HCF} \times \text{LCM} = \text{product of the two integers}$.
- Index laws: $a^m a^n = a^{m+n}$; $a^m / a^n = a^{m-n}$; $(a^m)^n = a^{mn}$; $a^0 = 1$ ($a \neq 0$); $a^{-m} = 1/a^m$; $a^{m/n} = \text{nth root of } a^m$ where defined.
- Standard form is $a \times 10^n$ with $1 \leq a < 10$.
- A simplified fraction has a terminating decimal only when its denominator has no prime factors other than 2 and/or 5.
- Recurring decimal to fraction: align the repeating block by multiplying by an appropriate power of 10, subtract to cancel the recurring tail, then simplify.
- Difference of squares: $a^2 - b^2 = (a-b)(a+b)$.
- Bounds from rounding: if x is rounded to the nearest unit u , then $x - u/2 \leq \text{true value} < x + u/2$.

RANKED SPEED TRICKS

1. Cancel before multiplying. For products of fractions, cross-cancel factors first. Large intermediate numbers are usually a sign that you missed a simplification.
2. Compare fractions by cross-products. For positive denominators, $a/b < c/d$ exactly when $ad < bc$. You often do not need common denominators or decimals.
3. Use factor pairs before prime-factor trees. For modest numbers, spotting an obvious common factor is faster than fully factorising both numbers.
4. For primality, stop at the square root. If n has a non-trivial factor, at least one factor is $\leq \sqrt{n}$. Test prime divisors only up to that point.
5. Use the complement in counting. For "at least two digits the same", total codes minus all-distinct codes is shorter than listing every repetition pattern.
6. Keep mixed numbers out of multiplication/division. Convert to improper fractions immediately, then cancel.
7. In a complex fraction, simplify structure first. Factor or cancel numerator and denominator before reaching for long division or decimals.
8. Separate coefficient and power of ten. $(4 \times 10^5)(8 \times 10^{-3}) \rightarrow (4 \times 8) \times 10^{5-3}$, then normalise.
9. Normalise only at the end. 32×10^2 is easier to obtain first; then convert to 3.2×10^3 .
10. For addition/subtraction in standard form, align exponents. $6 \times 10^4 + 8 \times 10^3 = (6 + 0.8) \times 10^4$.
11. Extract the largest square factor from a surd. $\sqrt{72} = \sqrt{36 \times 2} = 6\sqrt{2}$.
12. Rationalise with the conjugate. For denominators $a + \sqrt{b}$, multiply by $a - \sqrt{b}$ so the denominator becomes a difference of squares.
13. Recognise square-surds backwards. Expressions such as $A + B\sqrt{C}$ may be a square of $p + q\sqrt{C}$. Check the middle term $2pq\sqrt{C}$.
14. Do not decimalise exact values early. Fractions, π and surds are often easier to compare or simplify in exact form.

ADDITIONAL SPEED TRICKS — *new, not in the original guide*

1. Squaring numbers ending in 5: $(10a+5)^2 = 100 \cdot a \cdot (a+1) + 25$. $65^2 \rightarrow 6 \cdot 7 = 42 \rightarrow 4225$. Instant, and rarely automated even by strong students.
2. Near-square multiplication via difference of squares: $97 \cdot 103 = (100-3)(100+3) = 10000-9 = 9991$. Same identity M4 already flags for algebra — worth having as a numeric-arithmetic reflex too.
3. Bounds by monotonicity, not enumeration — for positive A and B , with the denominator bounded away from zero: the maximum of A/B is $UB(A)/LB(B)$ directly, without computing all four bound combinations and comparing them. Check the positivity/sign conditions hold before applying it; they are not automatic.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Expanding or multiplying large numbers before cancelling.
- Assuming $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$. It is not.
- Forgetting the negative power means reciprocal.
- Leaving a standard-form coefficient outside the range $1 \leq a < 10$.
- Using the rounded stated value itself as an upper or lower bound.
- Treating an estimate as if it were an exact result.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Can anything cancel?
- Can I stay exact?

- Is there a factorisation or power law?
- Is the answer in valid standard form?
- For bounds, did I use the correct endpoint combination?
- Does the magnitude agree with a rough estimate?

EXPANDED PATTERN BANK

Bounds, rounding and estimation

- Upper/lower bounds: for a positive product, upper x upper gives the upper bound and lower x lower gives the lower bound.
- Positive quotient: maximum numerator / minimum denominator gives the upper bound; minimum numerator / maximum denominator gives the lower bound.
- Strict upper boundary: a value rounded to a stated accuracy usually has an interval closed at the lower end and open at the upper end.
- Estimate aggressively: round to one or two significant figures only as far as needed to make mental arithmetic easy.
- Use nearby exact squares/cubes: $\sqrt{50}$ is a little above 7 because $49 = 7^2$. This can eliminate implausible options quickly.

M3. Ratio and proportion

Official guide: pp. 74-114 | Sections M3.1-M3.11

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Move fluently between ratio, fraction, scale factor, percentage multiplier and proportional equation. The strongest shortcut in this unit is choosing the representation that makes the relationship multiplicative rather than verbal.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Percentage multipliers and reverse percentages	Multipliers make repeated and reverse changes much shorter and safer.
AUTOMATE	Direct and inverse proportion	Translate the wording into $y = kx^n$ or $y = k/x^n$ immediately.
AUTOMATE	Scale factor, area and volume	Linear factor k implies area factor k^2 and volume factor k^3 .
AUTOMATE	Ratio-fraction conversion	For $a:b$, the fraction of the total corresponding to a is $a/(a+b)$.
FLUENT	Combining ratios and mixtures	Equalise the common quantity before merging ratios.
FLUENT	Growth, decay and compound interest	Repeated proportional change is one multiplier raised to a power.
TRAP-GUARD	Maps and simple sharing	Straightforward if units are made consistent first.

MEMORISE COLD

- Divide Q in ratio $a:b:c$ by first finding one part $= Q/(a+b+c)$.
- If $A:B = a:b$, then $A/B = a/b$. If the whole consists only of A and B , fraction $A = a/(a+b)$.
- Percentage multiplier: increase $r\% \rightarrow x(1 + r/100)$; decrease $r\% \rightarrow x(1 - r/100)$.
- Reverse a percentage change by dividing by the multiplier, not by applying the opposite percentage.
- Direct proportion: $y = kx^n$. Inverse proportion: $y = k/x^n$. Determine k from one known pair.
- Similarity: length ratio k ; area ratio k^2 ; volume ratio k^3 .
- Simple interest over n periods: interest = principal $\times$ rate $\times n$; the rate must be written as a decimal (or divide by 100 if given as a percentage).
- Repeated growth/decay: final = initial $\times$ multiplier ^{n} .

RANKED SPEED TRICKS

1. Convert ratio to fractions of the whole immediately. A mixture 3:7 means the first component is 3/10 of the total. This is often cleaner than continuing to speak in ratio language.
2. Use the unitary method when numbers are awkward. If 12 items cost £42, one item costs £3.50; scale from there. This is often faster than setting up cross-multiplication.
3. Combine ratios through the shared term. If $A:B = 2:3$ and $B:C = 4:5$, make B the same: 8:12 and 12:15, so $A:B:C = 8:12:15$.
4. Percentages are multipliers. 17% increase means $\times 1.17$. Keep the multiplier visible if more than one change occurs.
5. Reverse by division. If a final value is after a 20% increase, original = final/1.2. Subtracting 20% from the final is wrong.

6. Successive changes multiply. +20% then -20% gives $1.2 \times 0.8 = 0.96$, so the net change is -4%.
7. For proportion, find k once. Do not repeatedly use verbal scaling if one equation $y = kx^n$ captures the entire problem.
8. Recognise direct proportion from the graph condition. A straight line alone is not sufficient. Direct proportionality requires a straight line through the origin.
9. Go back to a linear factor before comparing areas/volumes. If areas are in ratio 25:49, lengths are 5:7. If volumes are 8:27, lengths are 2:3.
10. Write repeated change as a power immediately. Five periods of 3% growth is $\times 1.03^5$, not five separate additions of 3%.

ADDITIONAL SPEED TRICKS — new, not in the original guide

1. Difference-given ratio case: if $x:y = 3:5$ and $y-x = 14$, value per part = $14/(5-3) = 7$. The guide covers combining ratios via a shared term, but not this equally common "difference, not total, is given" variant.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Comparing ratios before converting quantities to the same units.
- Confusing part:part with part:whole.
- Undoing an increase by subtracting the same percentage.
- Adding percentage changes instead of multiplying their multipliers.
- Using k^2 for a volume or k^3 for an area.
- Calling a non-origin straight line a direct-proportion graph.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Same units?
- Part:part or part:whole?
- Can I use a multiplier?
- Direct/inverse: what is the power and what is k ?
- Similarity: am I working with length, area or volume?
- Repeated change: have I used an exponent?

EXPANDED PATTERN BANK

Mixtures and concentration: quickest framework

- Step 1: translate each ratio into a fraction of the mixture.
- Step 2: compute the amount of the component, not just the total mixture.
- Step 3: after mixing, add component amounts and total amounts separately.
- Step 4: convert the final component fraction back to the requested ratio or percentage only at the end.

M4. Algebra

Official guide: pp. 115-201 | Sections M4.1-M4.19

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Make method selection faster than manipulation. The guide covers notation, indices, substitution, expansion/factorisation, rational expressions, rearranging formulae, identities, coordinates, straight lines, graphs, gradients/areas, equations, quadratics, inequalities and sequences. The high-value skill is recognising structure before writing algebra.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Factorisation and algebraic simplification	The fastest route often starts by reducing structure rather than expanding it.
AUTOMATE	Quadratics	You need rapid switching between factorisation, completing the square, formula and graph interpretation.
AUTOMATE	Straight lines, gradients and graph interpretation	Coordinate and graph questions reward immediate recognition of slope, intercept and area/gradient meaning.
AUTOMATE	Equations and simultaneous equations	Substitution/elimination should be chosen by whichever removes a variable with least work.
AUTOMATE	Inequalities	The sign and boundary logic must be exact.
FLUENT	Rational algebra and rearranging formulae	Usually routine, but factorisation first can remove most of the work.
FLUENT	Sequences	First/second differences should immediately suggest linear/quadratic nth-term structure.
TRAP-GUARD	Notation, identities and coordinates	Foundational content; errors tend to be careless rather than conceptual.

MEMORISE COLD

- Difference of squares: $a^2 - b^2 = (a-b)(a+b)$.
- Straight line: $y = mx + c$; gradient $m = (\text{change in } y)/(\text{change in } x)$. Parallel lines have equal gradients; perpendicular non-vertical lines have gradients whose product is -1.
- Quadratic formula: $x = (-b \pm \sqrt{b^2 - 4ac})/(2a)$.
- Completing square for $x^2 + bx + c$: $(x + b/2)^2 + c - b^2/4$.
- For $ax^2 + bx + c$, axis of symmetry $x = -b/(2a)$.
- Recognise standard graph shapes on sight: linear, quadratic, cubic, reciprocal, exponential, and the basic sine/cosine/tangent graphs included in Mathematics 1.
- Linear sequence nth term: $an + b$, where a is the first difference.
- Quadratic sequence: second difference = $2a$ when nth term is $an^2 + bn + c$.
- Inequality rule: multiplying or dividing by a negative reverses the inequality sign.

RANKED SPEED TRICKS

1. Factor before expanding. Whenever you see a difference of squares, common factor, or factorable quadratic, test that route before multiplying brackets out.

2. For algebraic fractions, factor every numerator and denominator first. Only then cancel common factors. Cancelling terms across addition/subtraction is invalid.
3. Use substitution variables for disguised quadratics. If an equation contains x^6 and x^3 , set $u = x^3$ and solve the quadratic in u .
4. Choose the variable that is already isolated. In simultaneous equations, if $y = \dots$, substitute it rather than manufacturing elimination coefficients.
5. Use answer choices as information when appropriate. If candidate values are simple and direct substitution is faster than full symbolic solving, test them. This is a method-selection shortcut, not guesswork.
6. Keep negative substitutions bracketed. If $x = -2$, write $(-2)^2$. This prevents the common sign error between $(-2)^2$ and -2^2 .
7. Rearrange formulae by structure. Undo outer operations first; treat grouped expressions as single objects until the final step.
8. Translate graph words into operations. Distance-time gradient means speed; speed-time area means distance; a curved graph needs tangent gradient for instantaneous rate.
9. Do not over-read graphs. If only an approximate value is possible, do not manufacture spurious precision.
10. For a straight-line equation, expose $y = mx + c$. Once in this form, gradient and intercept are visible.
11. Equation through a point with known gradient. Use $y - y_1 = m(x - x_1)$ mentally even if you later rearrange to $y = mx + c$.
12. Area under a piecewise linear graph. Split into rectangles/triangles/trapezia rather than searching for one large formula.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Expanding a rational expression before looking for cancellation.
- Cancelling across a sum, e.g. $(x+2)/x$ cannot cancel the x .
- Using factorisation when the quadratic is not factorable over simple integers and wasting time searching indefinitely.
- Forgetting that a tangent gradient on a curve is local, not an average gradient over a large interval.
- Reversing an inequality unnecessarily, or failing to reverse it when dividing by a negative.
- Treating a graph sketch as exact numerical evidence.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Can I factor?
- What is the shortest quadratic method?
- Can I substitute rather than eliminate?
- What do gradient/area mean in this context?
- Inequality: any negative multiplier/divisor?
- Sequence: first or second differences?

EXPANDED PATTERN BANK

Quadratics: choose the shortest tool

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Factorise	Best when integer/rational factors are visible. Gives roots almost immediately.

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Complete the square	Best for the turning point, range, graph translation, or an expression designed around a square.
FLUENT	Quadratic formula	Reliable fallback when factorisation is not obvious or exact irrational roots are required.
FLUENT	Graph	Best for approximate solutions or interpreting roots/turning points in context.

- Recognition rule: if c has only a few factor pairs and b is modest, try factorisation for a few seconds before using the formula.
- Turning point rule: if the question asks minimum/maximum or range, completing the square is usually the direct language.
- Symmetry rule: the roots, when real, are symmetric about $x = -b/(2a)$. This can provide a quick check.
- Graph rule: x -intercepts are roots; the y -intercept is c ; the sign of a gives the opening direction.

Inequalities and sequences

- Linear inequalities: solve like an equation, but reverse the sign if multiplying/dividing by a negative.
- Two-variable regions: draw the boundary line, decide solid/dashed from $\leq$ versus $<$, then test one simple point such as $(0,0)$ if it is not on the boundary.
- Linear sequence: first difference d gives dn plus a constant. Fit $n=1$ to find the constant.
- Quadratic sequence: second difference D gives leading coefficient $D/2$. Subtract that an^2 sequence; the remainder is linear.

Method-selection drill - name the method before doing the algebra

- $x^2 - 49 \rightarrow$ difference of squares.
- $x^2 - 5x + 6 = 0 \rightarrow$ factorise.
- Find the turning point of $x^2 + 6x + 2 \rightarrow$ complete the square.
- Solve $x^2 + 3x - 1 = 0$ exactly $\rightarrow$ quadratic formula is the reliable direct route.
- Line perpendicular to $y = 4x + 1 \rightarrow$ gradient $-1/4$.
- Sequence 3, 8, 15, 24, ... $\rightarrow$ second differences are constant; use a quadratic n th term.
- If symbolic work is expanding rapidly, stop and ask whether factorisation, substitution or a graph interpretation was missed.

M4. Algebra - expanded recognition drill

Use these as instant triggers. The target is to identify the shortest valid route before writing more than one line.

P1 recognition: see the structure first

- Repeated powers: x^4, x^2 , constant $\rightarrow$ try $u=x^2$. x^6 and $x^3 \rightarrow$ try $u=x^3$. Treat the repeated expression as one object.
- Difference of squares: $A^2 - B^2 \rightarrow (A-B)(A+B)$, including disguised forms such as $49x^2 - 16$.
- Algebraic fraction: factor numerator and denominator before expanding or finding common denominators. Cancellation may remove most of the question.
- Quadratic: easy integer roots $\rightarrow$ factorise; turning point/shape $\rightarrow$ complete square; awkward exact roots $\rightarrow$ formula; root count only $\rightarrow$ discriminant.
- Straight lines: $y=mx+c$ exposes gradient and intercept immediately. Parallel $\rightarrow$ same m ; perpendicular $\rightarrow$ negative reciprocal where defined.

Ranked speed extensions

1. Stop once the requested information is known. If the question asks only whether roots exist, $D=b^2-4ac$ is enough. If it asks only which distance is greater, compare squares.

2. Substitute answer options when the algebra is uglier than the options. Multiple choice sometimes makes verification faster than derivation. Keep domain restrictions in view.
3. Preserve factorised form. A product exposes zeros and signs. Expanding too early often destroys useful structure.
4. For simultaneous equations, eliminate the cheapest variable. If one equation already gives $y=f(x)$, substitute it; if coefficients line up, eliminate.
5. Inequality sign discipline. Addition/subtraction preserve direction; multiplication/division by a negative reverses it. Do not divide by an expression of unknown sign without case control.
6. Graph interpretation can replace algebra. Roots are x-axis intersections; simultaneous solutions are curve intersections; gradient/area information may already answer the question.

Micro-drill: say the method before the answer

- $x^4-5x^2+4=0 \rightarrow$ disguised quadratic, $u=x^2$.
- $x^2+6x+2 \rightarrow$ complete square if vertex/minimum is wanted: $(x+3)^2-7$.
- $49x^2-16 \rightarrow$ difference of squares.
- $-2x>6 \rightarrow$ divide by a negative and reverse the sign.
- Line through two points $\rightarrow$ gradient first, then $y=mx+c$.
- Two possible answer values for $x \rightarrow$ test them directly if substitution is shorter than solving.

10-second check

- Can I factorise? Is it a disguised quadratic? Do I need the roots or only information about them? Can a graph or an answer-option substitution finish the job faster? Have I preserved signs, domains and exact forms?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

M5. Geometry

Official guide: pp. 202-300 | Sections M5.1-M5.19

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Convert the diagram into a dense set of constraints as quickly as possible. The official guide covers angle facts, quadrilaterals, congruence, transformations, Pythagoras, circles, coordinates, plans/elevations, bearings, mensuration, similarity, trigonometry and vectors. The main speed gain is annotation and theorem recognition.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Angle chasing and diagram annotation	Many problems collapse once equal sides, parallel lines, right angles and radii are marked.
AUTOMATE	Circle theorems	Recognition is much faster than re-deriving the geometry.
AUTOMATE	Pythagoras and right-triangle trigonometry	Core tools in 2D, 3D, coordinate and bearing problems.
AUTOMATE	Similarity and scale	Length/area/volume relationships are high-leverage and easy to confuse.
AUTOMATE	Vectors for geometric proof	The target is usually to show two vectors are equal or one is a scalar multiple of another.
FLUENT	Bearings and coordinate geometry	Reliable if you draw north lines and use displacement/Pythagoras.
FLUENT	Mensuration and sectors	Formulae are simple; the danger is using the wrong fraction of a circle or wrong perpendicular height.
TRAP-GUARD	Transformations, plans/elevations, terminology	Mostly recognition and careful description.

MEMORISE COLD

- Angles: around a point 360 degrees; on a straight line 180 degrees; vertically opposite equal; triangle sum 180 degrees; quadrilateral sum 360 degrees; polygon interior sum $(n-2)180$ degrees; exterior angles sum 360 degrees.
- Congruence tests: SSS, SAS, ASA, RHS. AAA gives similarity, not congruence.
- Pythagoras: $a^2 + b^2 = c^2$ in a right-angled triangle.
- Circle theorems: centre angle = 2 x circumference angle on the same arc; angle in a semicircle 90 degrees; same-segment angles equal; tangent-radius angle 90 degrees; alternate segment theorem; opposite angles of a cyclic quadrilateral sum to 180 degrees.
- Circle formulae: circumference $2\pi r = \pi d$; area πr^2 ; cylinder volume $\pi r^2 h$.
- Sector/arc: fraction of full circle = $\theta/360$.
- Coordinate midpoint: $((x_1+x_2)/2, (y_1+y_2)/2)$.
- Right prism volume = cross-sectional area x length.
- Similarity: length k , area k^2 , volume k^3 .
- Exact trig: $\sin 0=0$, $\sin 30=1/2$, $\sin 45=\sqrt{2}/2$, $\sin 60=\sqrt{3}/2$, $\sin 90=1$; cos values reverse; $\tan 0=0$, $\tan 30=1/\sqrt{3}$, $\tan 45=1$, $\tan 60=\sqrt{3}$.
- Transformations: reflection needs a mirror line; rotation needs centre, angle and direction; translation needs a vector; enlargement needs centre and scale factor (including fractional/negative factors where specified).

- Plans/elevations: plan = view from directly above; front/side elevations are orthographic views from those directions.
- The official Mathematics 1 guide does not require candidates to recall or use the sine or cosine rules.

RANKED SPEED TRICKS

1. Draw or redraw the diagram immediately. The guide repeatedly advocates a sketch for geometry/bearing questions. Accuracy of scale is less important than correct relationships.
2. Annotate every forced equality. Equal radii create isosceles triangles; parallel lines create corresponding/alternate angle equalities; tangents create right angles to radii.
3. Work backwards from the requested angle or length. Ask which theorem could produce the target, then search for the missing input to that theorem.
4. Add a construction only when it creates a known structure. A useful extra line often creates a triangle, diameter, radius or parallel relationship. Do not add lines randomly.
5. Exploit isosceles triangles aggressively. Two radii to points on a circle are equal. This frequently creates equal base angles without further calculation.
6. For polygon angles, prefer exterior angles when the shape is regular. Each exterior angle of a regular n -gon is $360/n$; this is usually faster than interior-sum calculations.
7. Compare squared lengths if only relative size matters. Avoid square roots when you merely need to decide which distance is longer or whether a triangle is right-angled.
8. 3D Pythagoras is usually two 2D Pythagoras steps. Find a face diagonal first, then use it in a second right-angled triangle.
9. Coordinates are displacement. From (x_1, y_1) to (x_2, y_2) , use Δx and Δy ; distance is $\sqrt{(\Delta x)^2 + (\Delta y)^2}$.
10. Bearings: new point, new north line. Bearings are measured clockwise from north at the point of origin. Draw the north line at every point where a bearing is specified.
11. Use exact trig before approximating. At 30, 45, 60 degrees, exact values often simplify immediately.
12. Reconstruct exact trig if memory slips. Use a 45-45-90 triangle and half an equilateral triangle. This is safer than guessing.
13. Choose SOH-CAH-TOA by the sides actually present. Label opposite, adjacent and hypotenuse relative to the stated angle before choosing the ratio.
14. Find the linear scale first. Even if the question gives an area or volume ratio, take the square/cube root to recover the length ratio before solving lengths.
15. Sector questions are "fraction of a circle" questions. Use $\theta/360$ times circumference or area. No separate conceptual formula is needed.
16. Keep perpendicular height distinct from sloping side. Triangle, parallelogram and trapezium areas require perpendicular height.
17. For vector proofs, write two routes to the same point. Equating route expressions often reveals the required result.
18. To prove parallelism with vectors, aim for scalar multiple. If vector $AB = k$ vector CD , the directions are parallel; sign of k gives same/opposite orientation.
19. To prove a midpoint, aim for equal vectors or equal position-vector averages. This is often shorter than coordinate geometry.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Trusting the visual scale of a diagram. Diagrams may not be to scale.
- Using AAA as a congruence condition.

- Using the wrong arc, sector or central angle, especially major versus minor.
- Measuring a bearing anticlockwise or from the wrong point.
- Using a sloping side as a perpendicular height.
- Using k^2 or k^3 in the wrong direction when moving between similar shapes.
- Invoking sine/cosine rules as memorised Mathematics 1 requirements. The official guide explicitly says they are not expected here.

UNDER TIME PRESSURE: 10-SECOND CHECK

- What equalities/right angles/parallel lines can I mark?
- Which theorem is triggered?
- Can I avoid a square root?
- Bearing: clockwise from which north line?
- Length/area/volume scale?
- Exact trig available?
- Vector proof: can I compare two routes or find a scalar multiple?

EXPANDED PATTERN BANK

Circle theorem recognition triggers

- Diameter visible -> test angle in a semicircle.
- Tangent visible -> draw/notice the radius to the point of contact; it is perpendicular.
- Same chord/arc feeding two circumference angles -> same-segment theorem.
- Centre and circumference angles subtend same arc -> centre angle is double.
- Four vertices on a circle -> cyclic quadrilateral; opposite angles sum to 180 degrees.
- Tangent plus chord plus circumference angle -> alternate segment theorem.

Geometry recognition drill - say the trigger, not the full solution

- Tangent meets radius at point of contact -> 90 degrees.
- Diameter subtends an angle at the circumference -> 90 degrees.
- Two radii form two sides of a triangle -> mark an isosceles triangle.
- Area ratio 49:81 for similar shapes -> length ratio 7:9.
- Volume ratio 125:216 for similar solids -> length ratio 5:6.
- Compare coordinate distances -> compare squared distances first if no exact length is required.
- Bearing 235 degrees from A -> draw north at A, then measure clockwise.
- Vector AB is -3 times vector CD -> the lines are parallel, with opposite directions.

M5. Geometry - expanded trigger bank

Geometry speed comes from annotating forced facts, then matching them to a theorem or construction.

P1 triggers to mark immediately

- Two radii in one triangle -> isosceles, so equal base angles.
- Diameter to a circumference point -> angle in a semicircle is 90 degrees.
- Tangent at a point on a circle -> radius to that point is perpendicular.
- Same chord/arc feeding two circumference angles -> equal angles in the same segment.
- Centre and circumference angles on the same arc -> centre angle is twice the circumference angle.
- Four points on one circle -> test cyclic quadrilateral; opposite angles sum to 180 degrees.
- Tangent plus chord -> test alternate segment theorem.

Ranked speed extensions

1. Work backwards from the requested quantity. Ask which theorem would produce it, then identify the one missing fact needed by that theorem.
2. Compare squared lengths before taking roots. This is faster for ordering distances and testing right angles.
3. Similarity: recover the linear factor first. Area ratio $\rightarrow$ square root; volume ratio $\rightarrow$ cube root; only then solve a length.
4. Bearings: new point, new north line. Measure clockwise from north at the point from which the bearing is stated.
5. Transformations require a complete description. Reflection line; rotation centre, angle and direction; translation vector; enlargement centre and scale factor.
6. Plans/elevations are projections, not perspective drawings. A plan fixes the horizontal footprint but does not by itself determine height.
7. Vector proof: two routes or scalar multiple. Equal route expressions prove a point relation; $AB=kCD$ proves parallelism (sign gives orientation).

Micro-drill: trigger only

- Area ratio 49:81 $\rightarrow$ length ratio 7:9.
- Volume ratio 125:216 $\rightarrow$ length ratio 5:6.
- A tangent touches at T and OT is a radius $\rightarrow$ angle OTB=90 degrees.
- A diameter subtends angle ACB $\rightarrow$ ACB=90 degrees.
- Need a regular n-gon exterior angle $\rightarrow$ $360/n$ is usually faster than interior-angle sums.
- Coordinate distance comparison only $\rightarrow$ compare $(\Delta x)^2 + (\Delta y)^2$; do not take square roots.

10-second check

- What can I mark equal or perpendicular? Is there a circle-theorem trigger? Is the diagram only illustrative, not to scale? Am I using length, area or volume scale? Is the height perpendicular? For a bearing, which point supplies the north line?

Expansion page: official syllabus/content basis retained; ranking and speed tactics are strategic synthesis.

M6. Statistics

Official guide: pp. 301-340 | Sections M6.1-M6.4

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Identify what the display is encoding before doing arithmetic. The official guide covers data displays, histograms, cumulative frequency, summary statistics and scatter graphs. The main risk for a strong student is not calculation; it is reading the wrong quantity from the representation.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Histograms and frequency density	Area represents frequency; unequal class widths make height alone misleading.
AUTOMATE	Cumulative frequency and quartiles	Read positions $N/4$, $N/2$ and $3N/4$ correctly and distinguish cumulative from class frequency.
AUTOMATE	Grouped-data estimates	Midpoints and interpolation create estimates, not exact values.
FLUENT	Scatter graphs and line of best fit	Correlation, interpolation/extrapolation and causation language must be precise.
FLUENT	Mean/median/mode/range comparisons	Choose like-for-like measures of centre and spread.
TRAP-GUARD	Basic charts and tables	Mostly straightforward display literacy and arithmetic.

MEMORISE COLD

- For an ordered list of n values, median position is $(n+1)/2$; if n is even, average the two middle values.
- Class boundaries from rounded data use the half-unit rule: a group 150-154 cm (nearest cm) corresponds to $149.5 \leq h < 154.5$.
- Histogram: frequency density = frequency / class width; frequency = area of bar = density x class width.
- Cumulative frequency quartile positions: lower quartile $N/4$; median $N/2$; upper quartile $3N/4$; IQR = $Q3 - Q1$.
- Grouped mean estimate = $\text{sum}(\text{midpoint} \times \text{frequency}) / \text{sum}(\text{frequency})$.
- Grouped values are estimates because the exact values within each class are unknown.
- Correlation does not imply causation.
- Interpolation is within the observed data range; extrapolation is outside it and is less reliable.

RANKED SPEED TRICKS

- Read axes, units and class widths first. Ten seconds spent here can prevent an entire wrong method.
- Histogram: think area, not height. If class widths differ, the tallest bar need not contain the greatest frequency.
- Convert cumulative frequencies to class frequencies by subtraction. The rise between successive class boundaries is the frequency in that interval.
- Mark $N/4$, $N/2$, $3N/4$ before touching the horizontal axis. This keeps quartile reading systematic.
- For a grouped mean, write midpoint x frequency directly. Do not reconstruct individual observations.
- Use an assumed mean mentally when values cluster. Work with deviations from a convenient central value, then add the mean back.

7. Compare centre and spread together. A higher median with a much larger IQR is not simply "better"; state both relevant differences.
8. Line of best fit: balance points. It should follow the overall trend, with roughly balanced scatter around it, rather than joining endpoints.
9. Do not overstate extrapolation. Outside the data range, the trend may change; treat the estimate as less secure.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Reading histogram height as frequency when class widths are unequal.
- Using cumulative frequency itself as a class frequency.
- Reporting a grouped-data mean/median as exact.
- Comparing a mean in one set with a median in another without justification.
- Calling correlation causal.
- Giving excessive precision from a hand-read graph.

UNDER TIME PRESSURE: 10-SECOND CHECK

- What does each axis represent?
- Equal class widths?
- Histogram: area or height?
- Cumulative: what is N?
- Exact data or grouped estimate?
- Am I interpolating or extrapolating?
- Correlation only, or is causation actually established?

EXPANDED PATTERN BANK

Display-selection triggers

- Categorical data -> bar chart, pie chart, pictogram, two-way/frequency table.
- Ungrouped discrete numerical data -> vertical line chart.
- Continuous/grouped numerical data -> histogram or cumulative frequency graph.
- Two numerical variables -> scatter graph.
- Values over time -> time-series line graph.

M7. Probability

Official guide: pp. 341-381 | Sections M7.1-M7.7

UNIT AIM FOR A KNOWLEDGEABLE STUDENT

Choose the representation before calculating. The guide covers experimental frequencies, fairness and expected outcomes, theoretical probability, exhaustive/mutually exclusive events, systematic enumeration, possibility spaces, addition/multiplication rules and conditional probability. The shortest solution is often a complement or a change of denominator.

PRIORITY RANKING

Tier	Focus	Why it matters for a knowledgeable student
AUTOMATE	Complements and exhaustive events	"At least one" and "not" questions often become one short subtraction from 1.
AUTOMATE	Tree diagrams: multiply along, add across	Sequential events become mechanical once dependency is handled correctly.
AUTOMATE	Conditional probability	The key conceptual move is to restrict the sample space and change the denominator.
AUTOMATE	Representation choice	Grid, Venn, tree, two-way table and frequency tree each simplify different structures.
FLUENT	Possibility spaces and equally likely outcomes	Counting is valid only when the elementary outcomes are equally likely.
FLUENT	Experimental versus theoretical probability	Relative frequency informs probability estimates but remains sample-dependent.
TRAP-GUARD	Basic probability scale and unknown probabilities	Routine if exhaustive/mutually exclusive structure is identified.

MEMORISE COLD

- $0 \leq P(A) \leq 1$; $P(A) + P(\text{not } A) = 1$.
- General OR rule: $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$. For mutually exclusive events the overlap is zero.
- Independent AND rule: $P(A \text{ and } B) = P(A)P(B)$.
- Conditional probability: $P(A \text{ given } B) = P(A \text{ and } B) / P(B)$, provided $P(B) > 0$.
- For exhaustive mutually exclusive outcomes, probabilities sum to 1.
- Sequential path probability: multiply conditional branch probabilities along the path.
- If several mutually exclusive paths produce the desired event, add their path probabilities.
- Independent events: the probability on a later branch is unchanged by the earlier result.
- Dependent events: later probabilities change after earlier outcomes, as in sampling without replacement.
- Conditional probability concept: $P(A \text{ given } B)$ uses only the B cases as the new sample space.
- Expected frequency = number of trials $\times$ probability, interpreted as a long-run expectation rather than a guaranteed count.

RANKED SPEED TRICKS

1. "At least one" $\rightarrow$ complement first. $P(\text{at least one success}) = 1 - P(\text{no successes})$. This can replace several cases with one path.

2. At every tree fork, branches sum to 1. Use this to fill missing probabilities immediately and to check the diagram.
3. Multiply along a route, add alternative routes. Keep these two operations conceptually separate.
4. Conditional probability = change the denominator. "Given B" means discard all non-B outcomes, then calculate the fraction of remaining cases that also satisfy A.
5. Without replacement: update both numerator and denominator. One object removed changes the total and possibly the favourable count.
6. Use counts instead of fractions when conditional structure is messy. A frequency tree or two-way table with convenient expected counts can make the same probability relationship much clearer.
7. Before counting outcomes, test equal likelihood. A sample-space count gives probability directly only if the elementary outcomes are equally likely.
8. Use symmetry. For fair dice, coins, cards or balanced possibility spaces, symmetric events can save enumeration.
9. For "A or B", check overlap. If A and B overlap, simple addition double-counts the intersection. Use inclusion-exclusion conceptually: add both, subtract overlap once.
10. Separate experimental evidence from theoretical certainty. A relative frequency close to a value is evidence for an estimate, not proof that the true probability equals it.

COMMON TIME-WASTERS AND TRAPS · TRAP-GUARD

- Confusing mutually exclusive with independent. Mutually exclusive non-zero-probability events are not independent.
- Adding probabilities of overlapping events without subtracting the overlap.
- Keeping the same branch probabilities after sampling without replacement.
- Using total sample size as the denominator in a conditional probability.
- Counting outcomes that are not equally likely as if each had the same probability.
- Treating expected frequency as a guaranteed observed frequency.

UNDER TIME PRESSURE: 10-SECOND CHECK

- Complement easier?
- Which representation?
- Independent or dependent?
- Tree fork sums to 1?
- Multiply along / add across?
- Conditional: what is the new denominator?
- Are elementary outcomes equally likely?

EXPANDED PATTERN BANK

Choose the representation first

- Small combined experiments with equally likely outcomes -> grid/possibility space.
- Sequential events -> tree diagram.
- Overlapping categories -> Venn diagram.
- Counts split by two categorical variables -> two-way table.
- A population split stage by stage -> frequency tree.

Conditional probability without formula memorisation

- Example structure: if 40 students study French and 10 of those also study Music, then among the French students the Music fraction is $10/40$. The denominator is 40 because "given French" has restricted the universe.
- Tree interpretation: a branch probability after an earlier event is already conditional on having reached that branch.
- Venn interpretation: given B means work only inside set B; numerator is the overlap A and B, denominator is all of B.

Final recall sheet

Final high-speed recall sheet

If you only rehearse 25 things

1. Cancel before multiplying.
2. Keep exact fractions/surds/pi exact until needed.
3. Estimate the order of magnitude before accepting a numerical answer.
4. Area conversion uses squared linear factor; volume uses cubed factor.
5. Percentage changes are multipliers; reverse by division.
6. Direct proportion must pass through the origin.
7. Similarity: length k , area k^2 , volume k^3 .
8. Repeated growth/decay is one multiplier to a power.
9. Factor before expanding.
10. Difference of squares should be instant.
11. Straight line $y=mx+c$; parallel same m ; perpendicular gradients multiply to -1 .
12. Choose the quadratic method from the question, not from habit.
13. Inequality sign reverses when multiplying/dividing by a negative.
14. Quadratic sequence: second difference = $2 \times$ coefficient of n^2 .
15. Geometry: redraw and annotate all forced equalities.
16. Two radii often create an isosceles triangle.
17. Circle theorem triggers: diameter, tangent, same chord, centre, cyclic quadrilateral.
18. Compare squared lengths when roots are unnecessary.
19. Bearings are clockwise from north at the stated point.
20. Exact trig values at $0, 30, 45, 60, 90$ degrees.
21. Histogram frequency is area, not necessarily height.
22. Cumulative-frequency quartiles are at $N/4$, $N/2$, $3N/4$.
23. "At least one" $\rightarrow$ try the complement.
24. Trees: multiply along, add mutually exclusive routes.
25. Conditional probability means restrict the sample space and change the denominator.

Last-pass test behaviour

- Read for structure before arithmetic. What type of object is this: ratio, factorisation, similar shapes, tree, histogram, quadratic, rate?
- Prefer reversible, checkable steps. Exact arithmetic and factored forms preserve information and make checking easier.
- Do not over-solve. If the question asks only which value is larger, a sign, a range or a ratio, stop when that information is determined.
- Use the answer as a checksum. Check sign, size, units, geometry plausibility, probability range and whether the value lies in the required domain.
- Abandon a bad route early. If the algebra is exploding, reconsider whether there was a cancellation, symmetry, complement or graph interpretation that you missed.

Source note

Official content basis: UAT-UK, ESAT Guide Mathematics 1, June 2026, pp. 1-381.

Unit page ranges used here: M1 pp. 5-12; M2 pp. 13-73; M3 pp. 74-114; M4 pp. 115-201; M5 pp. 202-300; M6 pp. 301-340; M7 pp. 341-381.

Priority labels and speed strategies are study recommendations derived from the structure and worked methods in the guide. They are not official statements of assessment weighting or future question frequency.